CHAPTER FOURTEEN

Relevant Aspects of Carbon and Low Alloy Steel Metallurgy

Contents

Wire Technology
ISBN 978-0-12-382092-1, DOI: 10.1016/B978-0-12-382092-1.00014-2

14.1. IMPORTANT PROPERTIES OF STEEL

Steel is the world's most used metal. Its role in engineering and technology is based on two major factors, cost and range of strength. Steel is relatively inexpensive, except for higher alloyed forms. Moreover, steel may be prepared and processed with strength ranging over two orders of magnitude, for example, from 30 MPa to over 3000 MPa. Steel is also available in ranges of toughness adequate for major structural applications.

These advantages are particularly pertinent for the case of *carbon steels*, which are simple alloys of iron and carbon, usually with small manganese additions. Processing flexibility can be increased with small additions of chromium, molybdenum, vanadium, and so forth. These alloys are called *low alloy steels*. This chapter will generally address carbon and low alloy steels.

A major disadvantage of steel is its vulnerability to corrosion ("rusting"). It lies below copper and copper alloys in the galvanic series. This vulnerability to corrosion is substantially mitigated with large chromium additions, and these alloys are called *stainless steels*. Stainless steels are discussed in Chapter 15 of this book. Of course, steel corrosion protection is provided by various coatings, and that subject is addressed in Chapter 16.

Performance in cutting and machine tool applications is greatly enhanced with large alloying additions of a number of elements, forming the basis for alloys called *tool steels*. Tool steels will be discussed in Chapter 15.

14.2. PRIMARY PROCESSING

The reservations expressed in Section 13.1 for copper processing also apply to steel. The processing of iron from its ores, and the making of steel are immense subjects, complicated by environmental considerations and the availability of scrap. For a comprehensive review of steel basics, the reader is referred to the *Making, Shaping and Treating of Steel*.[97]

14.2.1 Reduction of iron ore

Iron ores are basically iron oxides, and the first step in the overall production process for steel occurs in a *blast furnace*. The charge to the blast furnace consists of iron ore, coke (carbonaceous material), and limestone (mainly calcium carbonate). Preheated air is blown through the charge. Under steady-state operations, the following reactions occur. Near the air blast source, at the base of the furnace, coke burns completely, as per

$$C + O_2 \rightarrow CO_2 + \text{heat}. \quad (14.1)$$

Higher up in the furnace, carbon dioxide is reduced by hot coke, as per

$$CO_2 + C \rightarrow 2CO - \text{heat}. \quad (14.2)$$

Despite the loss of heat in Equation 14.2, a high temperature is retained and carbon monoxide reduces the iron ore. A representative equation is

$$Fe_2O_3 + 3CO \rightarrow 2Fe + 3CO_2. \quad (14.3)$$

As all of this is going on, waste material from the ore (silica, among many other substances) is converted into *slag* (silicates, etc.) using the following equations:

$$CaCO_3 \rightarrow CaO + CO_2; \quad (14.4)$$

$$2CaO + SiO_2 \rightarrow Ca_2SiO_4. \quad (14.5)$$

On a semi-continuous basis, iron ("pig iron") and slag are drawn off, and charge is added. The blast furnace may run continuously for years.

14.2.2 Steelmaking processes

The pig iron contains about 3 to 4.5 % carbon. Although these carbon levels are excellent precursors for *cast iron*, they are much too high for steel. Hence, steel must be processed for carbon reduction in "steelmaking" operations. Such operations have undergone many improvements over the last century and a half. Contemporary technologies are dominated by the *basic oxygen process* and by *electric arc processes*.

14.2.3 The basic oxygen process

This process involves the charging of molten pig iron and steel scrap into a vessel or "converter." A "lance" is inserted into the top of the converter, and oxygen is blown into the liquid. The oxygen reacts with the iron to form an iron oxide, which reacts with carbon, as per

$$FeO + C \rightarrow Fe + CO. \quad (14.6)$$

Lime, or other "slag-forming fluxes," are added. The overall reactions are self-sustaining, and carbon, manganese, silicon, sulfur, and phosphorous levels are rather quickly reduced, as shown in Figure 14.1.[98]

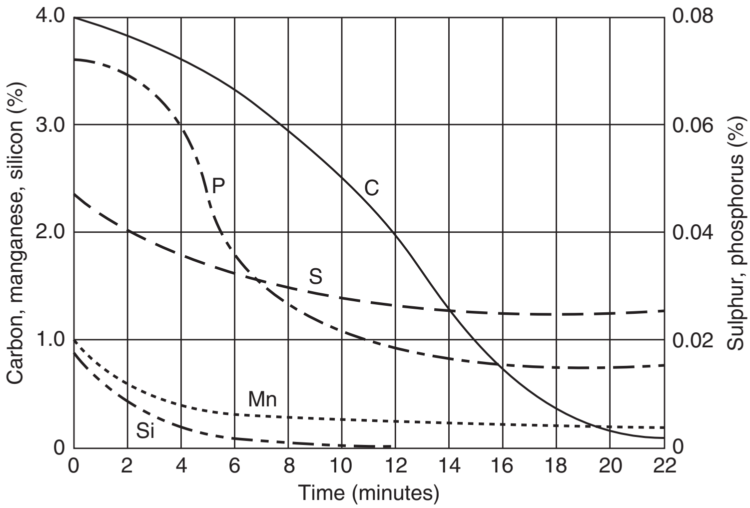

Figure 14.1 Reduction of carbon, phosphorous, sulfur, manganese, and silicon levels during basic oxygen steel making. From W. F. *Smith, Structure and Properties of Engineering Alloys*, Second Edition, McGraw-Hill, Inc., New York, 1993, 86. Copyright held by McGraw-Hill Education, New York, USA.

Variations on the basic oxygen process exist, including the blowing of oxygen from the bottom of the converter, and blowing a mixture of argon and nitrogen from the bottom, to provide stirring of the charge.

14.2.4 Electric-arc processing

In this technology, a cold charge of steel scrap is placed in a vessel, and an electric arc is struck between graphite electrodes and the scrap. The scrap is then melted. This process is especially useful where supplies of scrap are readily available. It also involves comparatively low capital cost, and provides good temperature control.

Electric arc processing may be necessary where easily oxidized elements such as chromium, tungsten, and molybdenum are to be added to alloy steels. Such alloy additions may be excessively oxidized in the oxygen-blowing practices of the basic oxygen process, resulting in costly losses of these additions. In the electric arc process, special slag covers protect against oxidation of alloying elements, as well as lowering phosphorous and sulfur levels.

14.2.5 Ladle metallurgy

Superior and more efficient refining can be achieved by transferring steel from basic-oxygen and electric-arc furnaces to a *ladle*. Ladle processing provides superior temperature and compositional homogenization in the molten steel. Precise deoxidation practices can be achieved with aluminum additions, and alloying additions can be made with increased precision. Cover slags can be employed to reduce sulfur and inclusions can also be floated into the slag and removed. The morphology of residual sulfide and oxide inclusions can be managed by additions of calcium and rare earths.

14.2.6 Vacuum degassing

Additional refinements can be achieved with ladle practice by way of *vacuum degassing*. In the most widely used process, a vacuum degassing system is placed over the ladle with two "legs" or "snorkels" extending into the molten steel. One snorkel introduces argon gas into the steel to force the metal into the degassing system through the other snorkel. Oxygen is introduced in the degassing system to reduce carbon in the steel with carbon monoxide formation. The vacuum degassed steel is returned to the ladle and a new cycle of degassing is initiated. Vacuum degassing practice can achieve very low carbon and hydrogen contents.

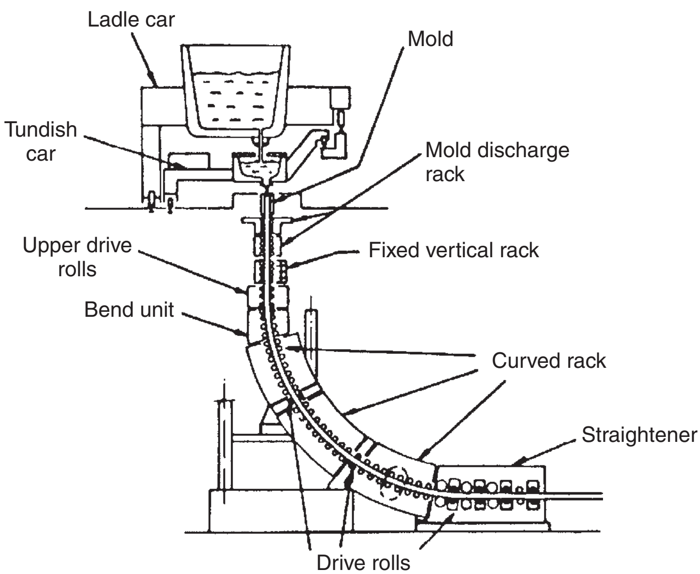

Figure 14.2 Schematic illustration of the continuous casting of steel. From W. F. *Smith, Structure and Properties of Engineering Alloys*, Second Edition, McGraw-Hill, Inc., New York, 1993, 89. Copyright held by McGraw-Hill Education, New York, USA.

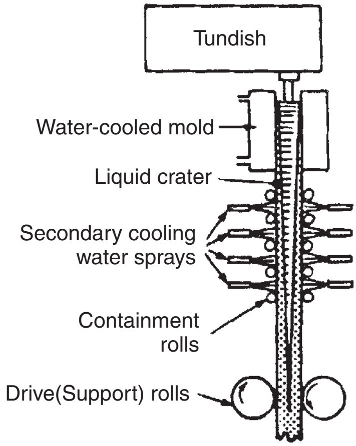

Figure 14.3 Schematic illustration of the mold region in Figure 14.2. From W. F. *Smith, Structure and Properties of Engineering Alloys*, Second Edition, McGraw-Hill, Inc., New York, 1993, 90. Copyright held by McGraw-Hill Education, New York, USA.

14.2.7 Casting practices

While individual ingots may be cast for certain steels, the majority of steel production is continuously cast. A detailed treatment of the continuous casting of steel is outside the scope of this book. However, Figures 14.2 and 14.3 provide illustrative details.[99,100] Rolling of the continuous cast product is discussed in Chapter 17.

14.3. THE IRON-IRON CARBIDE PHASE DIAGRAM

14.3.1 The phases of iron

The iron-rich end of the iron-iron carbide phase diagram is displayed in Figure 14.4.[101] Despite the relative complexity of this diagram, it is a good and necessary place to start the discussion of carbon steel. The left-hand side of the diagram represents *pure iron*. The left-hand side of the diagram, illustrates that up to 912°C iron exists as a phase called *ferrite*, or α iron. From 910 to 1400°C, iron exists as a phase called *austenite*, or γ iron. Finally, between 1400°C and the melting point at 1535°C, iron exists as a phase called *delta ferrite*, or δ iron.

The right-hand side of the iron-iron carbide phase diagram involves the phase iron carbide, or Fe_3C, often called *cementite*. Cementite is a hard, brittle material with a complex crystal structure. The carbon in Fe_3C is not

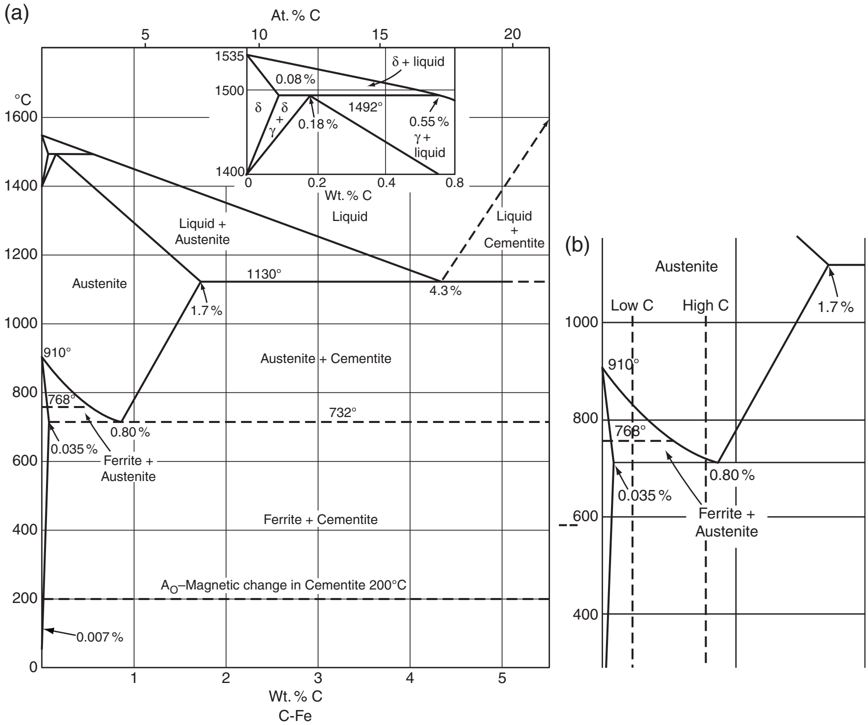

Figure 14.4 (a) Iron-rich end of the iron-iron carbide phase diagram, (b) enlarged portion of (a). From *Metals Reference Book*, C. J. Smithells, editor, Butterworths, London, UK, 1976, 510. Copyright held by Elsevier Limited, Oxford, UK.

as stable as pure carbon or graphite in the iron-carbon system. Thus, strictly speaking Figure 14.4 is not a stable phase diagram. However, for practical purposes it will be considered stable for this discussion. It should be noted that the portion of the phase diagram between ferrite and iron carbide, and below 732°C, consists of a two-phase combination of ferrite and cementite.

14.3.2 Crystal structure and the solubility of carbon

Now ferrite and delta ferrite have a body-centered-cubic (BCC) structure, as shown in Figure 13.13, and shown again for convenience in Figure 14.5. These two forms of iron can largely be regarded as the same phase, albeit separated by a temperature gap. The austenitic phase has a face-centered-cubic (FCC) structure, as shown in Figure 11.10, and again for convenience in Figure 14.6. A major difference between these two crystalline forms of

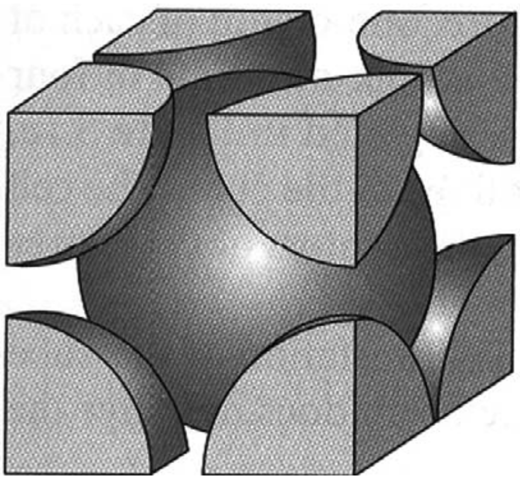

Figure 14.5 The BCC structure, as per ferrite and delta ferrite.

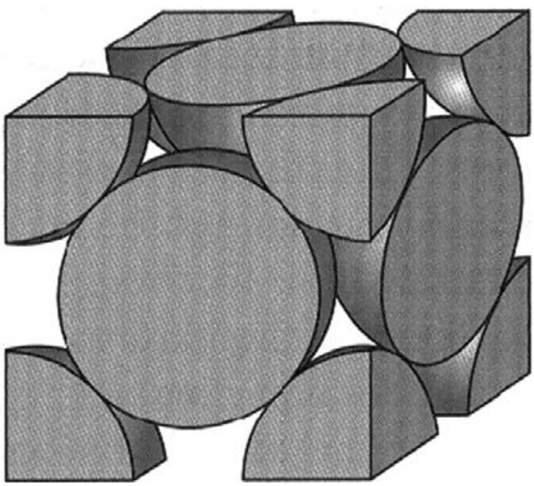

Figure 14.6 The FCC structure, as per austenite.

iron is their ability to accommodate carbon as a solute. Austenite has spacious locations for carbon atoms at the cell center, and at the centers of the cell edges, as illustrated in Figure 14.7. On the phase diagram it is clear that at 1130°C austenite can contain 1.7% carbon as solute, well above the carbon levels of steels. Ferrite, while less dense than austenite, has no such locations for carbon solute, and the maximum solubility of carbon in α ferrite is only 0.035% at 732°C. At carbon levels beyond this small solubility limit, the stable phases are ferrite (nearly pure iron) and cementite (or austenite).

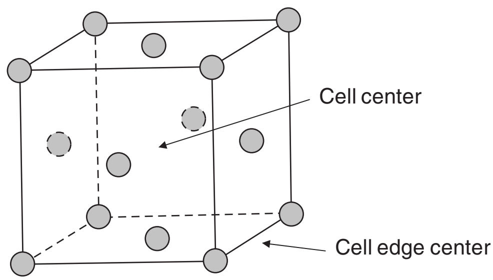

Figure 14.7 Spacious locations for carbon atoms in the FCC structure.

14.4. AUSTENITE DECOMPOSITION

14.4.1 Transformation of austenite upon cooling

The significance of the points discussed in Section 14.3.2 is that the majority of carbon steel processing scenarios involve, at one or more stages, heating the steel to the austenite range and cooling the steel below 732°C, thus precipitating some combination of ferrite and iron carbide. This 732°C transformation temperature is called the *eutectoid* temperature. A range of possibilities for these *austenite decomposition products* is shown in Figure 14.8.

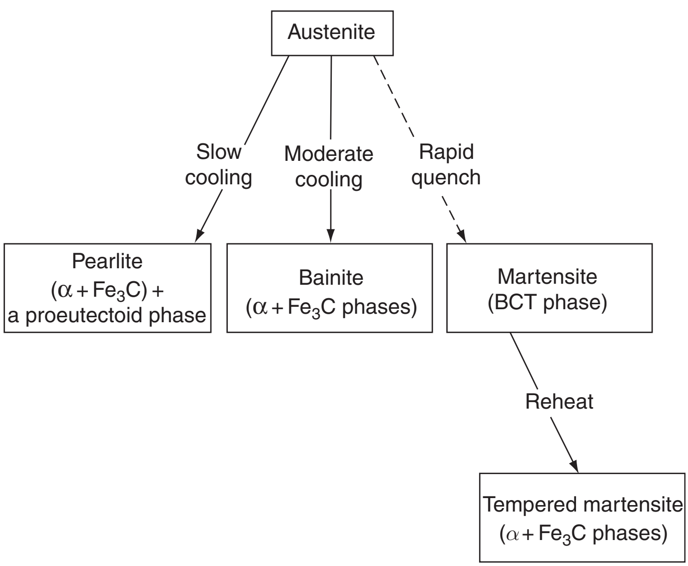

Figure 14.8 Possibilities for austenite decomposition products.

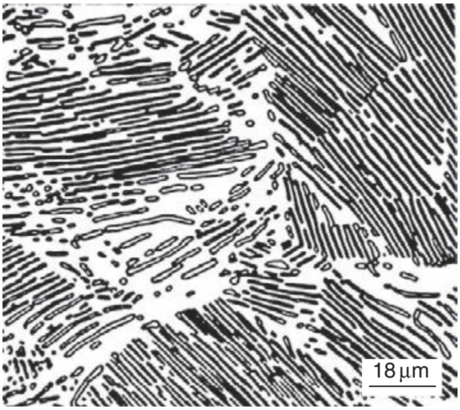

Figure 14.9 Fully pearlitic structure showing dark-etching layers as iron carbide and the light-etching layers as ferrite. From W. D. Callister, Jr. and D. G. Rethwisch, *Fundamentals of Materials Science and Engineering*, John Wiley & Sons, New York, 2008, 385. Reprinted with permission of John Wiley & Sons, Inc. Copyright held by John Wiley & Sons, Inc.

The leftmost arrow in Figure 14.8 notes that, upon slow cooling, austenite will transform to a structure involving a two-phase mixture of ferrite and cementite. These two phases are juxtaposed in a layered structure called pearlite. Pearlite consists of two phases, and the combination is properly called a "microconstituent." The characteristic pearlitic microstructure is shown in Figure 14.9.[102] Now, unless the steel has a composition of 0.80% carbon (the eutectoid composition), some bulk ferrite or carbide will appear along with the pearlite. This additional material is referred to as *proeutectoid*, since it forms above the eutectoid temperature. Figure 14.10 shows a microstructure containing a combination of proeutectoid ferrite and pearlite.[103] Steels that are heated into the austenite region and slowly cooled to form a proeutectoid phase and pearlite are said to have been *normalized*.

The center arrow in Figure 14.8 indicates that upon moderate cooling the austenite will transform to a ferrite plus cementite microconstituent called *bainite*. Bainite involves a very fine dispersion of cementite particles in a fine-grained "feathery" appearing ferrite structure. If the austenite is cooled rapidly enough, or quenched, the austenite will not form a combination of ferrite and cementite, but will dynamically transform to a body-centered structure with carbon atoms trapped similar to solute atoms. This phase is called *martensite*, and is, in effect, ferrite supersaturated with carbon. Martensite is unstable, and does not appear on the phase diagram in Figure 14.4.

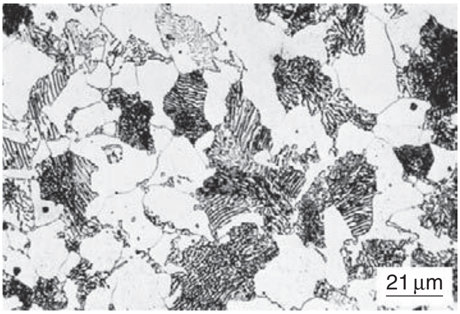

Figure 14.10 Microstructure with comparable amounts of pearlite and ferrite with ferrite as the light continuous phase. From W. D. Callister, Jr. and D. G. Rethwisch, *Fundamentals of Materials Science and Engineering*, John Wiley & Sons, New York, 2008, 387. Reprinted with permission of John Wiley & Sons, Inc. Copyright held by John Wiley & Sons, Inc.

14.4.2 Tempered martensite

Martensite, although hard, is quite brittle and impractical for most uses. However, its toughness can be greatly improved by *tempering*. This process involves heating the martensite to a temperature in the 250 to 650°C range. In this process, represented at the lower right in Figure 14.8, fine carbides precipitate from the martensite, leading to a ferrite-cementite structure similar to that of bainite. Figure 14.11 displays the effects of tempering temperature on the mechanical properties of tempered martensite.[104]

14.4.3 Spheroidite

Actually, none of the austenite transformation products shown in Figure 14.8 is stable, and if the transformation products are held for many hours at a temperature just below 732°C, the cementite will emerge as a globular or spherical phase in a ferrite matrix. This structure is called *spheroidite*, and a typical microstructure is shown in Figure 14.12.

Thermal treatment below the eutectoid temperature of 732°C is generally referred to as an *anneal*.

14.5. STRUCTURE-MECHANICAL PROPERTY RELATIONS

All else equal, increased carbon content increases the strength of steel. However, the microstructure plays an important role as well. These points are illustrated in Figure 14.13.[105] A general listing of typical tensile strengths for various steel microstructures and process histories is given in Table 14.1.

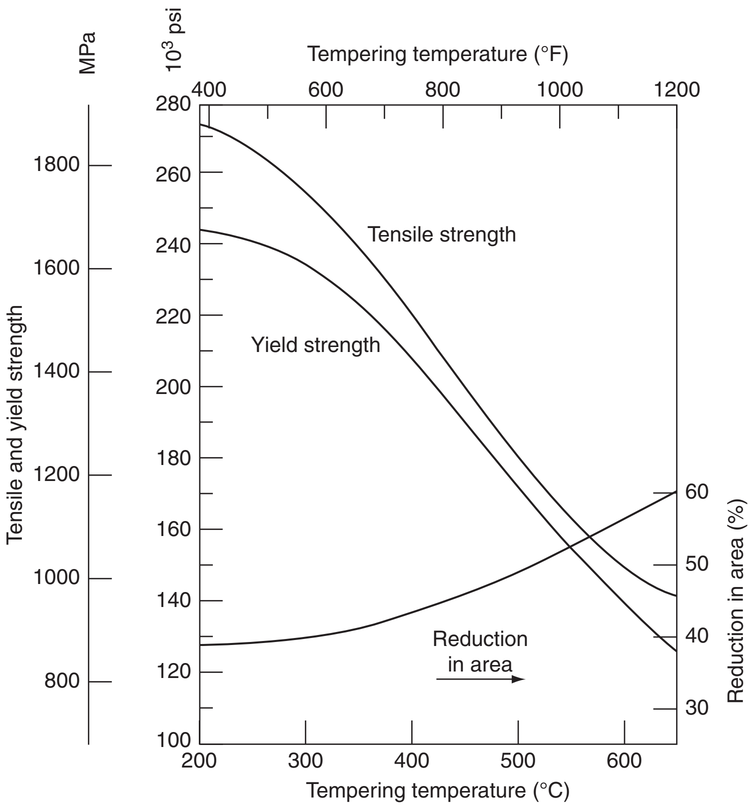

Figure 14.11 The effects of tempering on the mechanical properties of martensite. From W. D. Callister, Jr. and D. G. Rethwisch, *Fundamentals of Materials Science and Engineering*, John Wiley & Sons, New York, 2008, 436. Reprinted with permission of John Wiley & Sons, Inc. Copyright held by John Wiley & Sons, Inc.

14.6. TRANSFORMATION DIAGRAMS

14.6.1 Isothermal transformations

The transformation of austenite to pearlite does not happen instantaneously, and the time required varies depending on the point of cooling below the 732° C transformation temperature. This behavior is quantified in Figure 14.14.[106] Figure 14.14 is called an *isothermal transformation diagram*, and it represents the behavior of a eutectoid steel, or a steel with 0.80% carbon. As an example, suppose that the steel is very rapidly cooled from 800 to 600°C. It then goes from a temperature at which austenite is stable to a temperature at which austenite still exists, but is unstable. The region of austenite existence in the

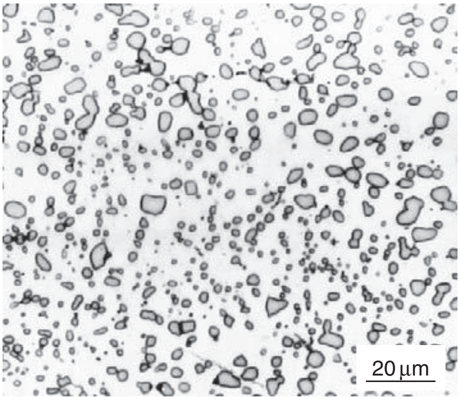

Figure 14.12 Spheroidized microstructure with globular carbides in a ferrite matrix. From W. D. Callister, Jr. and D. G. Rethwisch, *Fundamentals of Materials Science and Engineering*, John Wiley & Sons, New York, 2008, 420. Reprinted with permission of John Wiley & Sons, Inc. Copyright held by John Wiley & Sons, Inc.

transformation diagram is labeled A. Now, if the steel is held isothermally at 550°C, it will start to transform to pearlite in a little less than one second, and the transformation to pearlite will be complete in about seven seconds.

The transformation illustrated in Figure 14.14 is very important because it creates a very-fine-grained pearlite structure, which possesses excellent drawability, en route to very high strength. It is the basis for the manufacture of tire cord and other ultra-high-strength steel wire products. Isothermal processing to achieve this very-fine-grained pearlite is called *patenting*. Historically, the isothermal condition has been achieved by quenching into molten lead, held at the desired transformation temperature.

If the transformation occurred at 650°C, a coarser pearlite structure would have developed with lower strength potential and less drawability. An isothermal transformation at, about 400°C would have produced bainite.

In general, higher carbon steels have a structure that is primarily pearlite under drawing conditions, and low carbon steels are drawn with a structure that is primarily ferrite, consistent with the vertical lines in Figure 14.4. All structures become stronger with increased drawing reduction, and an unworked structure can be restored with annealing or normalizing.

14.6.2 Continuous cooling (non-isothermal) transformations

While isothermal processing is certainly industrially feasible and allows for rather precise process control, it is often expeditious to use non-isothermal, continuous cooling transformation instead. Figure 14.15 displays the

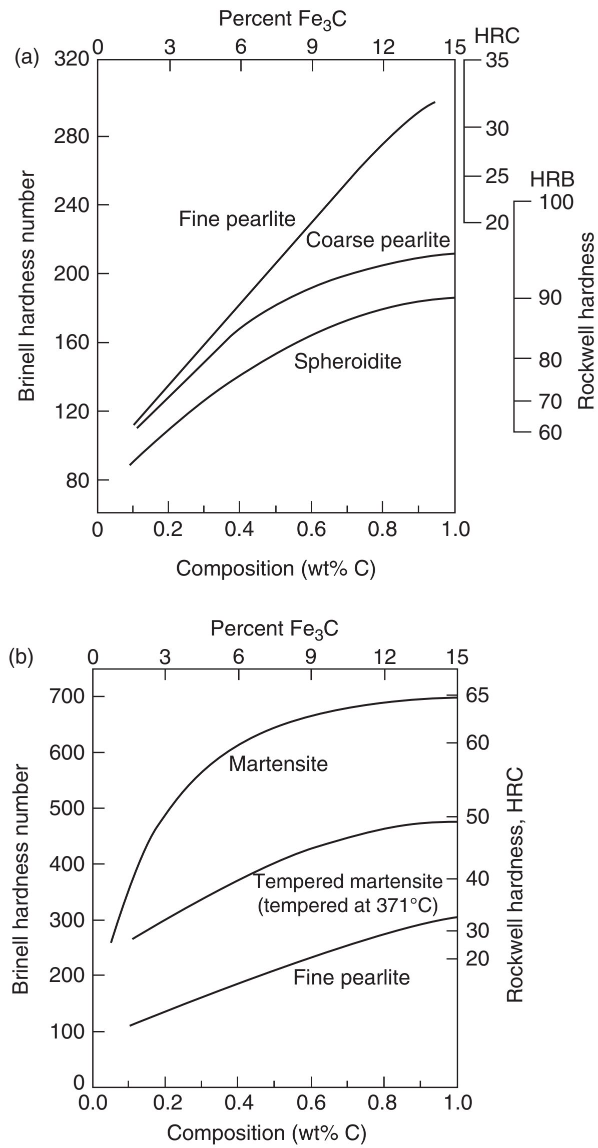

Figure 14.13 Effect of carbon content and microstructure on the hardness of steel. From W. D. Callister, Jr. and D. G. Rethwisch, *Fundamentals of Materials Science and Engineering*, John Wiley & Sons, New York, 2008, 432–433. Reprinted with permission of John Wiley & Sons, Inc. Copyright held by John Wiley & Sons, Inc.

Table 14.1 Typical mechanical properties for a variety of steel compositions and structures

Composition, Microstructure, Process History	Tensile Strength
Very low C (0.06%), ferrite, normalized	360–405 MPa
Medium C (0.30%), ferrite/pearlite, normalized	640–720
Medium C (0.30%), ferrite/pearlite, cold drawn	1000–1500
High C (0.7%), pearlite, normalized	1050–1125
High C (0.7%), pearlite, cold drawn	2000–3000
Medium C (0.40%), bainite	1500–2000
Medium C (0.40%), tempered martensite	1000–2000
Medium C (0.40%), spheroidite	400–500

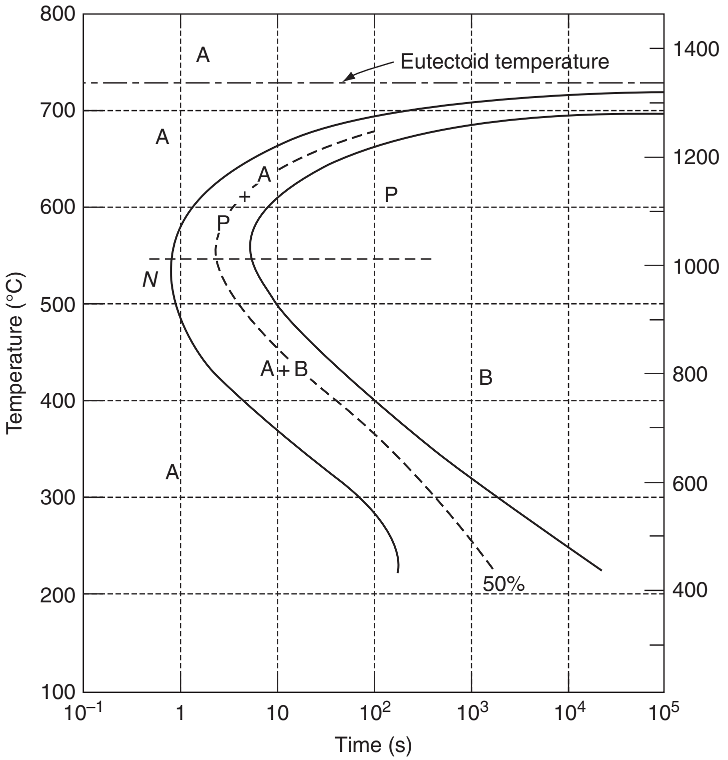

Figure 14.14 Isothermal transformation diagram for a eutectoid steel (0.76% carbon). From W. D. Callister, Jr. and D. G. Rethwisch, *Fundamentals of Materials Science and Engineering*, John Wiley & Sons, New York, 2008, 420. Reprinted with permission of John Wiley & Sons, Inc. Copyright held by John Wiley & Sons, Inc.

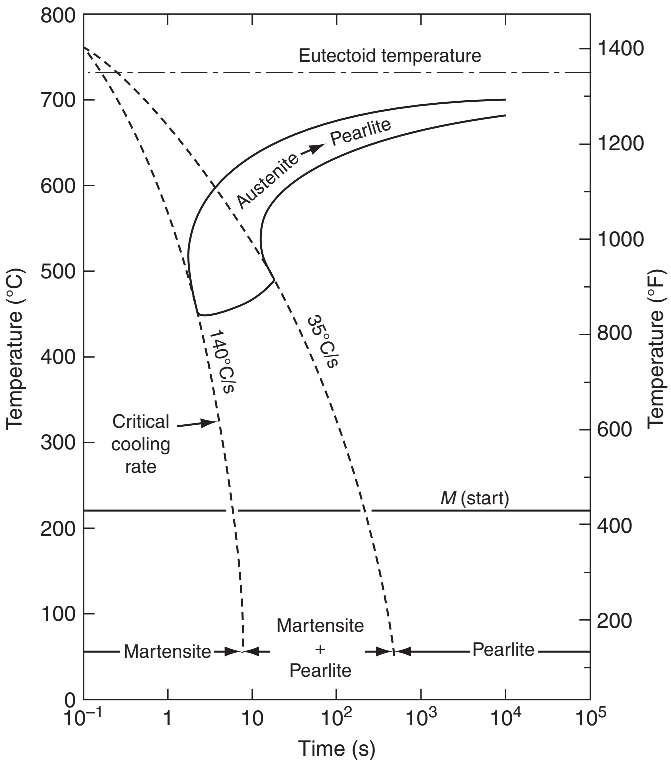

Figure 14.15 Transformation diagram for continuous cooling responses for a eutectoid steel (0.76% carbon). From W. D. Callister, Jr. and D. G. Rethwisch, *Fundamentals of Materials Science and Engineering*, John Wiley & Sons, New York, 2008, 429. Reprinted with permission of John Wiley & Sons, Inc. Copyright held by John Wiley & Sons, Inc.

transformation responses for continuous cooling for a eutectoid steel.[107] The Stelmor® process uses such an approach, developing pearlitic microstructures for increased drawability or increased descaling response.[108]

14.6.3 Forming martensite

Cooling more rapidly than the critical cooling rate of 140°C indicated in Figure 14.15 involves cooling rates that avoid the austenite-pearlite transformation region. Thus, the pearlite transformation is entirely suppressed. Once the temperature reaches the M_s temperature at approximately 220°C, the austenite will start to transform, nearly instantaneously, into martensite. This transformation is not time dependent, but it is temperature dependent; that is, as the temperature gets lower, more martensite forms. Eventually, at a temperature referred to as M_f, the transformation to martensite should be complete. The level of the M_f may be somewhat obscure; however, and martensite transformation temperatures are often identified with the degree of martensite transformation. For example, M_{90} would indicate the temperature at which 90% of the martensitic transformation has occurred (roughly 100°C for a eutectoid steel).

Alloying has a substantial effect on the curves in Figure 14.15. In nearly all cases, alloying lowers the martensite transformation temperatures. Alloying also delays the transformation to pearlite, or displaces the "noses" or the "C-curves," representing pearlite transformation, to higher times. This latter effect can be very useful in the thermal processing of steels, since slower, more easily achieved cooling rates can result in avoidance of pearlite, and full transformation to martensite can occur. Steels with slower pearlite transformation and with C-curve noses displaced to longer times are said to be more *hardenable*, and the alloying additions that accomplish this end are said to increase *hardenability*.

The effects of alloying elements in lowering the martensite transformation temperature and in delaying pearlite transformation do not occur for cobalt. Moreover, at higher levels increased carbon will not improve hardenability.

14.7. FLOW STRESS, COLD WORKING, AND ANNEALING

14.7.1 The flow stress of steel

The strength of steel, as previously discussed, varies enormously with composition and microstructure. Nonetheless, for cold working purposes, annealed and normalized steel strength data can, as with other metals, be fitted to and projected from the equation $\sigma_o = k\varepsilon_o^N$. This will generally require laboratory testing dedicated to the properties of the steel of interest.

Consider the following examples. Figure 14.16 displays an extraordinary range of flow stress versus strain relations for very low carbon iron (0.007% C).[109] The stress-strain relations can be approximated by the relation:

$$\sigma_o(\text{MPa}) = 430\varepsilon_o^{0.59}. \tag{14.7}$$

The stress-strain relationship for 1020 steel (0.20% C) in Figure 11.8 can be approximated by

$$\sigma_o(\text{MPa}) = 725\varepsilon_o^{0.40}. \tag{14.8}$$

The stress-strain relationship for 0.27% C steel in Figure 14.17[37] can be approximated by

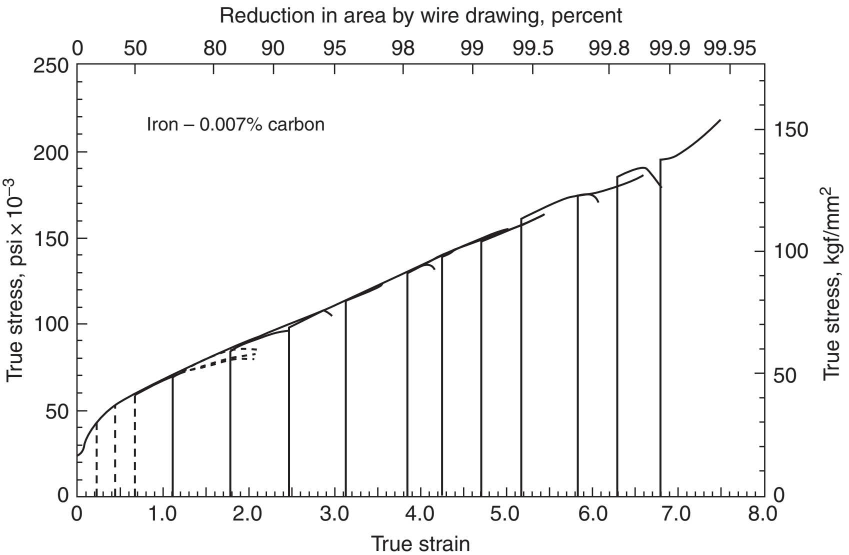

Figure 14.16 Flow stress for 0.007% carbon iron. From G. Lankford and M. Cohen, *Transactions of the American Society for Metals*, 62 (1969) 623. Copyright held by ASM International, Materials Park, OH.

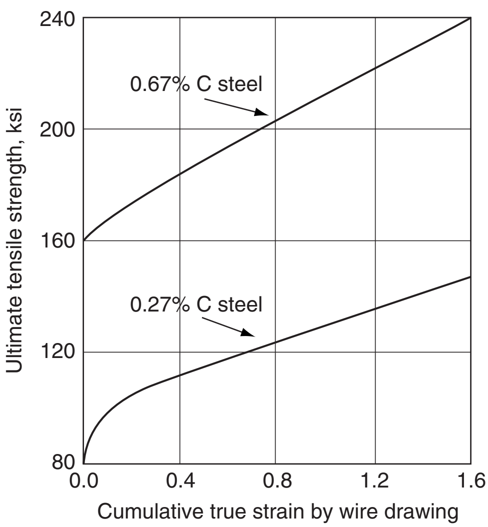

Figure 14.17 Tensile strength versus true strain curves for 0.27 and 0.67% carbon steel wires, drawn with successive 20% reductions. Data taken from A. B. Dove, Deformation in Cold Drawing and Its Effects, *Steel Wire Handbook*, Vol. 2, The Wire Association, Inc., Branford, CT, 1969, 16.

$$\sigma_o(\text{MPa}) = 960\varepsilon_o^{0.23}. \tag{14.9}$$

Finally, the stress-strain relationship for 0.67% C steel in Figure 14.17 can be approximated by

$$\sigma_o(\text{MPa}) = 1485\varepsilon_o^{0.19}. \tag{14.10}$$

There is a trend in these data-fitting equations for N to decrease and for k to increase with increased carbon content. However, such generalizations should not be a substitute for running a few tests on the material of interest. The approach illustrated in Figures 14.16 and 13.3 (pulling tensile tests after each drawing pass and superimposing the curves to make a master curve) is the recommended method.

14.7.2 Annealing cold worked steel

Understanding the annealing procedures for cold worked steel requires another look at the iron-iron carbide phase diagram in Figure 14.4. Annealing with the simple objective or removing cold work is generally called *process annealing*. It is undertaken *below* the eutectoid temperature (732°C) so austenite and its transformation products are not inadvertently formed. However, temperatures as high as 700°C can be used. A nominal time of one hour, at temperature, is typical of such anneals. Extended time anneals just below the eutectoid temperature are used to establish spheroidized structure, as noted in Section 14.4.3.

Where relief of residual stresses is desired, without necessarily recrystallizing the structure, anneals at 550°C or lower are used. Again a nominal time of one hour at temperature is typical.

It was noted in Section 14.4.1 that heating of steel into the austenite region, followed by relatively slow cooling, is called *normalizing*. Another thermal treatment that should be noted, involving heating above the eutectoid temperature, is the *full anneal*. This process involves heating at about 50°C above the temperature at which the steel becomes nearly all austenite, and then cooling very slowly ("furnace cooling"). This procedure forms relatively soft, coarse pearlite, which may be advantageous for certain machining or forming operations.

14.8. AGING IN STEEL

The behavior discussed previously is often complicated by the relocation of carbon (and nitrogen) atoms over time at ambient temperature, and over very short times at moderately elevated temperature. This topic was first mentioned in Section 11.2.5, and Figure 11.7 describes the yield point phenomenon, which reflects the role of *aging* in steel. A comprehensive discussion of this subject follows. The author has published a practical, wire-industry-focused review of aging in steel.[110]

14.8.1 Introduction

As a metallurgical term, aging usually refers to an increase in strength and hardness, coupled with a decrease in ductility and toughness that occurs over time, at a given temperature. In some cases these changes achieve a maximum or minimum at a certain time with conditions at later times referred to as "overaged."

Such aging is usually attributable to changes in the state or location of solute elements. There are two major modes: aging that involves precipitation from a supersaturated solid solution is referred to as *quench aging*, and *strain aging* involves the redistribution of solute elements to the strain fields of dislocations. When strain aging occurs in the absence of concurrent plastic deformation it is called *static*, and when strain aging occurs concurrently with plastic deformation it is called *dynamic*. Generally, the rate of aging increases with temperature, unless the solubility of the solute element changes substantially. In the temperature range of dynamic strain aging, the aging response is quite rapid. It should be noted that quench aging and strain aging may coexist.

In steel, the practical aging responses involve the role of carbon and nitrogen in ferrite (or in martensite, if tempering is viewed as an aging process). There are many nuances of this type of aging in steel wire processing and related mechanical properties. Strain aging, in particular, is pertinent to this chapter.

14.8.2 Static stain aging in steel

Strain aging in steel involves the positioning of carbon and nitrogen atoms that are actually in solution in ferrite. As shown in Table 14.2, the population of these atoms should be quite low at ambient temperature, especially for the case of carbon.[111] Nonetheless, the phenomenon of strain aging is marked, even at ambient temperature.

Basically, any interstitial solute carbon or nitrogen atom exerts a certain *compression* on the surrounding crystal structure. Moreover, dislocation stress fields involve regions of *tension*. Thus, the mechanical energy of the steel can be lowered if carbon and nitrogen atoms migrate into the regions of tension associated with the dislocations. Since energy is

Table 14.2 Approximate temperatures for 0.01, 0.001, and 0.0001% solubility limits of iron carbide and nitrides in ferrite

Compound	0.010 (°C)	0.001 (°C)	0.0001 (°C)
Fe_3C	600	450	300
Fe_4N	350	200	100
$Fe_{16}N_2$	200	100	—

From *The Making, Shaping and Treating of Steel*, 10th Edition, W. T. Lankford, Jr., N. T. Samways, R. F. Craven and H. E. McGannon (Eds.), Association of Iron and Steel Engineers, Pittsburgh, PA, 1985, 1284. With permission.

lowered, this migration occurs naturally with time. However, the dislocations must now increase the mechanical energy of the steel if they are to move away from the preferentially located carbon and nitrogen atoms. This requires extra stress for dislocation motion, and means that the steel has a higher yield strength. The increase in yield strength associated with strain aging is manifested in an upper yield point, as shown in Figure 14.18 at about 240 MPa.[112] The data in Figure 14.18 are from a "box-annealed" steel, where the slow cooling ensures a pronounced strain aging. As the tensile stress reaches the upper yield point, dislocations

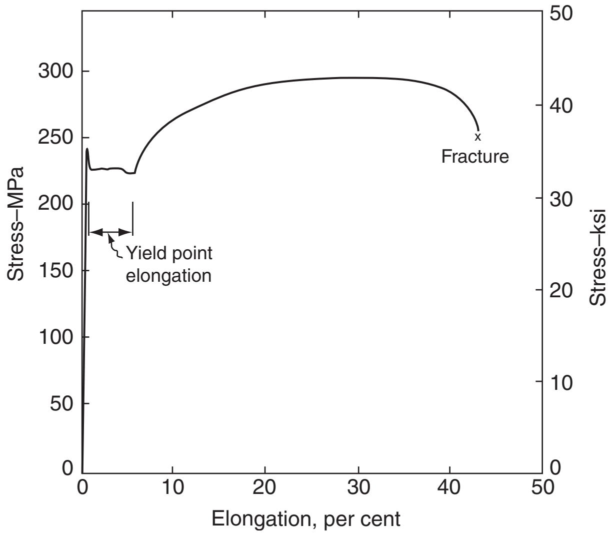

Figure 14.18 Stress-strain curve for "box annealed" carbon steel. *The Making, Shaping and Treating of Steel*, 10th Edition, W. T. Lankford, Jr., N. T. Samways, R. F. Craven and H. E. McGannon (Eds.), Association of Iron and Steel Engineers, Pittsburgh, PA, 1985, 1400. Copyright held by Association for Iron & Steel Technology, Warrendale, PA, USA.

presumably are able to move away from their "atmospheres" of carbon and nitrogen. The moving dislocations multiply, as well, and a large population of unencumbered dislocations allows deformation to occur more readily, resulting in a drop in stress to the lower yield point shown in Figure 14.18 at about 225 MPa. This shift occurs in stages, involving Lüders bands in the steel, as noted in Figure 11.7b. Eventually the yield point elongation is achieved and normal strain hardening ensues. The magnitude of the upper yield point depends on the stiffness of the testing machine and the strain rate of the tensile test.

The yield point phenomenon can be a major nuisance in many forming operations (resulting in Lüders lines or stretcher strains, "fluting," discontinuous yielding, etc.). The yield point can be removed by deforming the steel to just beyond the yield point elongation. This is especially common with sheet and strip processing for deep drawing purposes, for which a light "temper" rolling pass, or "skin pass", is used, as illustrated in Figure 14.19. In wire processing, a light sizing pass or a straightening pass may accomplish the same objective. It is essential to remember, however, that strain aging will not stop, and within a relatively short time the yield point will be observed again; that is, there will be a time window for any yield-point-free forming to be accomplished.

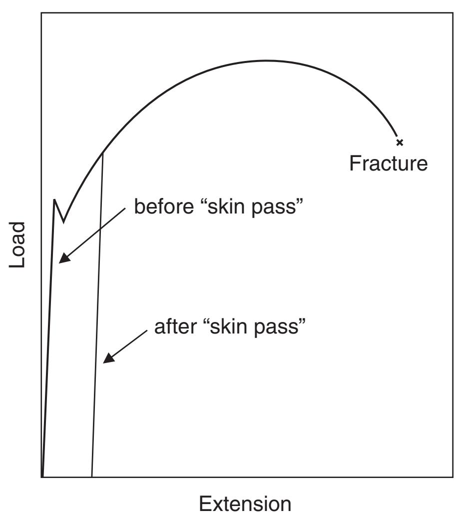

Figure 14.19 Illustration of the change in load-elongation curve with a light "skin pass."

Table 14.3 Approximate times for full strain aging in carbon steels at different temperatures

Temperature (°C)	Time
0°C	20 wks
21	10 wks
100	1½ hrs
120	20 min
150	4 min

The time for strain aging is a strong function of temperature. The degree of strain aging represented in Figure 14.18 can be achieved in about 1½ hours at 100°C, and Table 14.3 provides some time-temperature relationships for substantial aging. In this context, it is important to note that strain aging does not continue into an overaged mode, and, as full strain age strengthening is achieved, no subsequent decrease in strength can be expected with time.

14.8.3 Dynamic strain aging in steel

Extrapolation of the data of Table 14.3 to higher temperatures indicates that times can be so short they involve strain aging concurrent with plastic deformation, even at practical processing and tensile testing strain rates. When strain aging occurs concurrently with plastic deformation it is called dynamic.

Dynamic strain aging in steel, at processing strain rates, occurs most noticeably in the 200 to 650°C range, as displayed in Figure 14.20.[113] The flow stress or strength data for a strain rate of $6.6 \times 10^{-2}s^{-1}$ shows a marked peak at about 270°C, and the strength data for strain rates of $10s^{-1}$ and $430s^{-1}$ show peaks at about 400°C and 500°C, respectively. These peaks result from the inhibition of dislocation motion by strain aging from carbon and nitrogen atoms, and the carbon and nitrogen are so mobile that moving dislocations cannot pull away. The decline in strength with temperature above the peaks in Figure 14.20 may reflect the onset of recovery mechanisms.

The impact of strain rate on the peak positions in Figure 14.20 creates a region of *anomalous strain rate effect* in the 250°C range; that is, in most cases increased strain rate leads to increased strength, but in this region it is the opposite. In the anomalous strain rate region increased strain rate produces faster average velocities for dislocations and/or increased numbers of

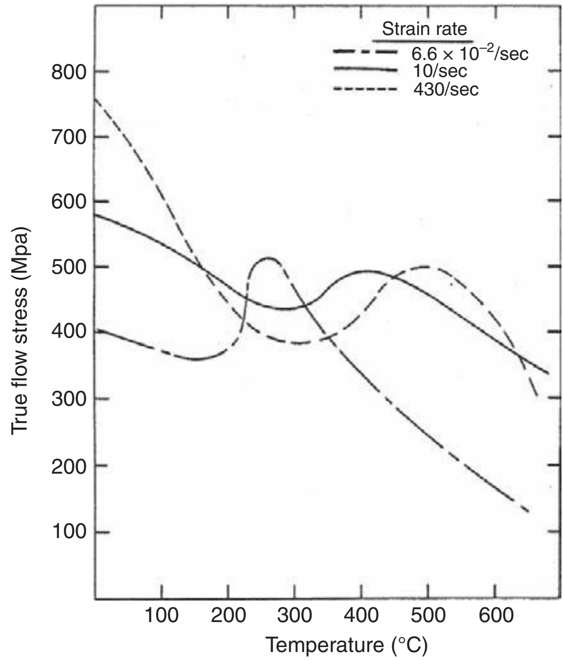

Figure 14.20 Stress-strain rate-temperature relations for a low carbon steel displaying the anomalous strain rate effect due to dynamic aging. From M. E. Donnelly, The Flow Stress of Low Carbon Steels Under Hot Deformation: A Basic Data Compilation for Process Analysis, M. S. Project, Rensselaer Polytechnic Institute, 1982.

dislocations in such a way that the strain aging effects of carbon and nitrogen cannot keep up. Hence the steel has *decreased strength*, not increased strength, at the higher strain rate.

14.8.4 Relevance to Pearlite-Containing Steels

The previous discussions have focused on the state of carbon and nitrogen in ferrite alone, and it is in ferrite that the aging processes occur. The dominance of such behavior in very low carbon steel or spheroidized steel is obvious. However, ferrite is also critical to the mechanical behavior of pearlite-containing steels, and such steels display the aging responses. This is not surprising, since fully pearlitic, eutectoid steels are nearly 90% ferrite.

14.8.5 Implications for wire processing

First, all carbon steel wire conditions are prone to static strain aging, to the consequent appearance of yield point phenomena, fluting (flattening in bending), and discontinuous yielding in forming operations. Moreover, any carbon steel wire that has been in service more than a few months at ambient temperature may be assumed to be fully strain aged.

Dynamic strain aging is a major consideration in steel wire processing, since thermomechanical heating and limited cooling capability can easily result in deformation temperatures of 200°C and higher. Moreover, such temperatures are apt to be unstable and result in varying degrees of dynamic strain aging. Much effort has been devoted to this, including highly engineered cooling systems, modeling and measurement of average temperatures during drawing, tapered drafting schedules, and so forth. It is important to note how ASTM Designation A 648-04a, "Steel Wire Hard Drawn for Prestressing Concrete Pipe," addresses the matter of wire temperature and strain aging. Subsection 4.3 of Designation A 648-04a is quoted below.

> *4.3 The wire shall be cold drawn to produce the desired mechanical properties. The wire manufacturer shall take dependable precautions during wire drawing to preclude detrimental strain aging of the wire.*
> *Note 2 — Allowing wire to remain at elevated temperatures, such as 400°F (204°C) for more than 5 s or 360°F (182°C) for more than 20 s, can result in detrimental strain aging of the wire. Detrimentally strain aged wire typically has reduced ductility and increased susceptibility to hydrogen embrittlement.*

Designation A 648-04a refers to strain aging in general, and lists times at temperature that would be relevant to *static* strain aging. However, the implications are that wire drawing is occurring at higher temperatures than temperatures that would be pertinent to 5 or 20 s duration.

The temperatures for dynamic strain aging in steel wire drawing are dependent on strain rate. The strain rate in wire drawing is discussed in Section 7.2.2. The strain rate range of 10^3 to 10^4 s^{-1} corresponds with commercial steel drawing practice. On the basis of the data in Figure 14.20, the indicated dynamic strain aging range would be about 500 to 650°C. While this temperature range is exceptionally high for the average drawing temperature in sophisticated steel drawing practice, it is not unreasonable for the wire surface temperature, as discussed in Section 6.1.7. Such temperatures only exist for a short time, but time is not a variable in the case of dynamic strain aging.

The ASTM Designation A 648-04a guidelines have emerged from carefully developed empirical databases, and are clearly a basis for careful practice. They likely minimize immediate static strain aging, and may require that dynamic strain aging conditions be avoided for the bulk of the wire cross section during drawing. However, it seems that the surface of the wire may have been dynamically strain aged in any case.

In analysis of practical situations, the dynamic strain aging of the wire surface must be considered in conjunction with such issues as lubricant breakdown, crow's feet development, oxidation, and even austenitization (and subsequent martensite development). Compromised wire properties from excessive drawing temperature cannot, in these contexts, be attributed solely to strain aging.

It is especially important, however, to note that lower strain rate drawing, forming, fastener manufacturing, and so forth, need not reach the 500 to 650°C range for dynamic strain aging to occur. Indeed, temperatures in the 200 to 400°C range will allow dynamic strain aging conditions to occur in operations of modest deformation rate (Figure 14.20).

Strain aging is actually usefully promoted by so-called "low temperature" stress-relieving heat treatments. Technically, a ferrous stress-relieving heat treatment should be a thermal exposure below the austenite range that relieves internal stresses by allowing dislocation motion over time. The dislocations are driven by the internal stresses and move with time, producing creep. The creep strains relieve the driving stresses, and in time the stresses are substantially relaxed. Classical carbon steel stress relieving has involved such treatments as one hour at 540°C, and so forth. In the steel wire industry, it has been common practice to use the term stress relieving for heat treatments as low as 200°C. Such heat treatments maximize strength and minimize stress relaxation in cold drawn wire. However, it should be pointed out that this property improvement almost certainly results from the rapid static strain aging that occurs with these heat treatments. It is difficult to believe that much dislocation motion occurs at such a low temperature, especially in the context of dislocation immobilization by way of static strain aging. In any event, the low temperature stress relief heat treatment can be beneficial where stress relaxation is a problem in fasteners, cables, reinforcing wires, and other products that bear high sustained stresses.

14.8.6 Control of aging

Development of aging in the short term can be minimized by avoiding some of the contributing processing conditions noted earlier. The introduction of a light temper or straightening pass will temporarily remove the effects of aging, as illustrated in Figure 14.19. Strain aging will soon return, however, as shown in Table 14.3.

From a compositional or alloy perspective, it is impractical to reduce aging simply by lowering carbon and nitrogen. However, aging can be greatly reduced by tying up nitrogen and carbon as stable nitrides and carbides. For example, as little as 0.05 w/o aluminum can, with proper processing, tie up nitrogen as AlN, and titanium and niobium additions have long been used to tie up carbon.

14.9. CARBON STEEL COMPOSITIONS

A generally useful system of steel designation is the Unified Numbering System (UNS), which incorporates the traditional Society of Automotive Engineers (SAE) and American Iron and Steel Institute (AISI) systems. For carbon steels, the UNS designation starts with the letter "G," followed by the four numbers of the SAE-AISI system, usually followed by 0. There are many international steel designations, as well, but they can usually be correlated with the UNS system. To simplify notation, this text uses the four numbers of the SAE-AISI system, or 10XX for most carbon steels, where the XX designation indicates the weight percent carbon in hundredths of a percent. For example, plain carbon steel with 0.20 weight percent carbon content would have an SAE-AISI designation of 1020 and a UNS designation of G10200.

Plain carbon steel chemical specifications usually call out only four elements (other than iron): carbon, manganese, sulfur, and phosphorous. Carbon increases strength and, up to intermediate levels, increases hardenability. Manganese also increases strength and hardenability, and is beneficial to surface quality in most cases. Increased carbon and manganese can compromise ductility and weldability, however. Phosphorous and sulfur generally decrease ductility and toughness, an especially big problem in the early days of steel making, and their chemical specifications involve maximum tolerances in the case of 10XX plain carbon steels. As an example, the specification for SAE-AISI 1020 or UNS G10200 steel is as follows (with compositions in weight per cent): 0.18– 0.23% carbon, 0.30–0.60% manganese, 0.040% maximum phosphorous, and 0.050% maximum sulfur.

There are three other carbon steel specification categories, 11XX, 12XX, and 15XX, using the SAE-AISI system. The 15XX system involves increased manganese additions. The 11XX "free cutting" or re-sulfurized system involves sulfur typically in the range of 0.10%, where the sulfur greatly improves machinability through the effect of finely dispersed sulfides. Finally, phosphorous also improves machinability, and the 13XX system involves both sulfur and phosphorous, and sometimes lead, additions for that purpose.

It is important to understand that carbon steels contain numerous additional "residual" elements in their composition. In general these elements are not specified, even when it is understood that their levels are not small. Of significant interest is silicon, which is retained mostly from its role in steelmaking deoxidation practice. Levels of residual silicon can range from negligible to as much as 0.60%. Other important residual elements can be traced to pig iron and, especially, steel scrap used in primary processing of steel. Such elements include chromium, nickel, copper, molybdenum, aluminum, vanadium, and so on. The sum total of the residual elements can be the order of 1.00% in carbon steels. It is generally assumed that these elements are "harmless" or even "beneficial." Certainly they generally increase strength. However, their impact on ductility, toughness, weldability, and surface quality may be problematical.

14.10. LOW ALLOY STEEL COMPOSITIONS

The majority of desirable steel property combinations can be developed with plain carbon steels, provided there is adequate processing capability or limited product dimensions. However, plain carbon steels that are to be hardened (by martensite formation) require relatively rapid cooling, which may not be practical for the interior of a larger cross section. Moreover, such rapid cooling may involve the risk of distortion or cracking. Alloying generally improves hardenability, or allows slower cooling to be undertaken en route to the formation of martensite. A great improvement in hardenability can be achieved with minimal alloying (a very few percent of alloying additions), and many low alloy steels or simply alloy steels, have been designed with this in mind. There are numerous categories of lightly alloyed steels addressing objectives such as corrosion resistance and electromagnetic response; however, this discussion will focus on alloying for hardenability and general mechanical property improvement.

Manganese is generally present in low alloy steels, as it is in plain carbon steels, and its role in improving hardenability has been cited. Chromium is perhaps the most prominent alloying addition to low alloy steels. Chromium greatly improves hardenability, and its carbides are hard and stable at higher temperatures, thus expanding the temperature range for the use of the steel. The SAE-AISI designation for chromium low alloy steels is 50XX or 51XX, where the XX designation indicates the carbon content, in hundredths of a percent.

Molybdenum is highly effective in increasing the hardenability of low alloy steels. Moreover, it largely eliminates the reduced toughness ("temper embrittlement") often observed in low alloy steels upon tempering exposures in the 250–400°C range. The SAE-AISI designation for molybdenum steels is 40XX. Chromium and molybdenum additions are often used together ("chrome-moly" steels), and the relevant SAE-AISI designation is 41XX.

Nickel additions generally add to the robustness of low alloy steels in that nickel inhibits some of the grain coarsening tendencies associated with chromium additions, and generally leads to increased toughness, or a lowering of the ductile-to-brittle transition temperature. Chromium-nickel-molybdenum low alloy steels are preferred for many applications, and have the relevant SAE-AISI designations of 43XX and 86XX, among others.

Other alloying elements are often employed in improving the processing latitude and properties of hardenable low alloy steels, particularly vanadium and boron. In addition to increasing hardenability, vanadium refines grain size and forms strong, stable carbides. The relevant SAE-AISI designation for chromium-vanadium low alloy steels is 61XX. Boron can be extremely potent in promoting hardenability, with a common range of 0.0005–0.003 weight percent. Boron steels are generally designated by placing a B before the number indicating carbon content, such as 86B45 indicating a chromium-nickel-molybdenum low alloy steel containing a boron addition and about 0.45% carbon.

14.11. QUESTIONS AND PROBLEMS

14.11.1 Figure 14.11 shows the effects of tempering on the mechanical properties of an alloy steel. Can this steel be tempered to manifest at least 45% area reduction at fracture with a tensile strength minimum of 1150 MPa?

Answer: Yes, tempering below 525°C will ensure a tensile strength of at least 1150 MPa, and tempering above 440°C will ensure an area reduction of at least 45%. Thus, a tempering treatment between 440 and 525°C should work.

14.11.2 Consider Figure 14.14 and an isothermal treatment at 650°C. When will pearlite start to form, and when will pearlite formation be complete?

Answer: Placing a line at 650°C, and recognizing that the time scale is logarithmic, it appears that pearlite will start to form at about 8 seconds and finish forming at about 80 seconds.

14.11.3 What appears to be the dependence of uniform elongation on carbon content in carbon steels?

Answer: One approach to answering this question is to realize that when stress-strain relations are expressed as $\sigma_o = k\varepsilon_o^N$, it can be shown that the uniform elongation actually equals N (note Equation 11.11). Now looking at Equations 14.7 through 14.10, it is clear that respective carbon levels of 0.007%, 0.20%, 0.27%, and 0.67% correspond to respective uniform elongation levels of 0.59, 0.40, 0.23, and 0.19. Thus, uniform elongation decreases as carbon content increases.

14.11.4 Table 14.2 displays three solubility limits for Fe_3C in iron. Using the phase diagram of Figure 14.4, add a fourth solubility limit.

Answer: Near the lower left-hand corner of the diagram it is clear that a carbon content solubility limit of 0.035% is associated with ferrite at 732°C.